

Valence Alignment Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Statistical Contrasts

PENPAL Project

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1 Overview

This notebook performs comparative analysis across all three experimental conditions:

- **Human-AI:** Human co-writing with an LLM
- **Human-Human:** Two humans co-writing together
- **AI-AI:** Two AI agents co-writing (simulated baseline)

Key Questions: 1. Do alignment patterns differ across the three conditions? 2. Is alignment asymmetric in all conditions or condition-specific? 3. Does the AI-AI baseline show the strongest alignment (as expected)? 4. How do effect sizes compare across conditions?

2 Load Data

```
## Data availability:

## # A tibble: 6 x 4
##   file                                `human-ai` `human-human` `ai-ai`
##   <chr>                                <lgl>      <lgl>      <lgl>
## 1 dyadic_sentiment_scores.parquet TRUE       TRUE       TRUE
## 2 sentiment_scores.parquet        FALSE      FALSE      FALSE
## 3 surface_metrics.parquet         TRUE       FALSE      FALSE
## 4 semantic_exploration.parquet    FALSE      FALSE      FALSE
## 5 embeddings.parquet              FALSE      FALSE      FALSE
## 6 cleaned_stories.csv             FALSE      FALSE      FALSE

##
## Loaded data summary:

## # A tibble: 3 x 3
##   condition  n_stories n_turns
##   <fct>      <int>   <int>
## 1 Human-AI      87     870
## 2 Human-Human   38     380
## 3 AI-AI         40     400
```

3 Preprocessing

```
## Analysis dataset: 1485 turns across 165 stories
```

3.1 Build Long-Format Directional Dataset

Following the within-condition analysis approach: we stack both alignment directions into a single long-format dataset. This enables proper statistical testing of directional asymmetry via predictor $\times$ direction interactions.

- **A1 $\rightarrow$ A2** (same-turn): Author 2 responds to Author 1's current valence
- **A2 $\rightarrow$ A1** (cross-turn): Author 1 responds to Author 2's previous valence

```
## === Long-Format Alignment Data ===

## # A tibble: 6 x 6
##   condition  alignment_direction n_rows n_stories mean_predictor mean_response
##   <fct>      <fct>              <int>   <int>         <dbl>         <dbl>
## 1 Human-AI   A1_to_A2              783     87         -0.0831         0.00386
## 2 Human-AI   A2_to_A1              783     87          0.0173        -0.0831
## 3 Human-Human A1_to_A2              342     38         -0.210         -0.172
## 4 Human-Human A2_to_A1              342     38         -0.178         -0.210
## 5 AI-AI      A1_to_A2              360     40          0.0877         0.0734
## 6 AI-AI      A2_to_A1              360     40          0.0739         0.0877
```

3.2 Standardize Within Conversation $\times$ Direction

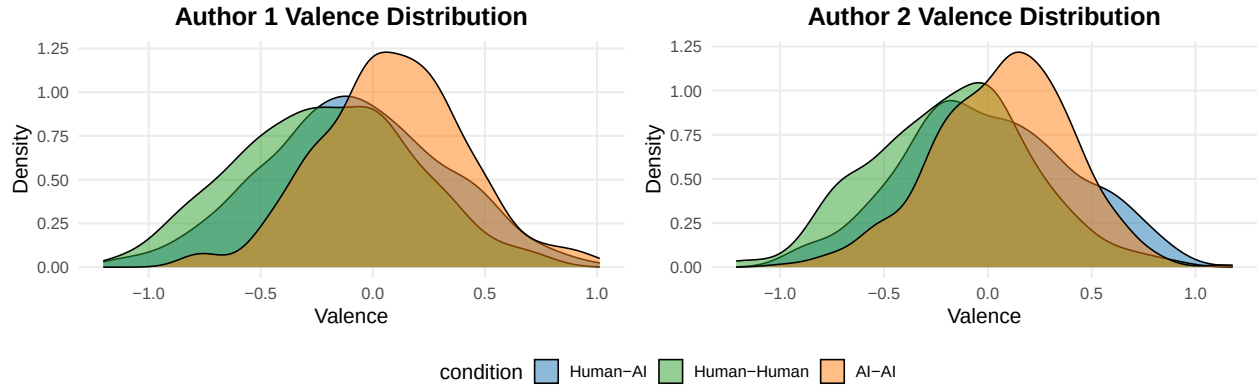
Z-score predictor and response valence within each (conversation_id $\times$ direction) cell. This ensures alignment coefficients are comparable across directions and conditions.

```
## Standardized dataset: 2970 observations
```

4 Descriptive Statistics

Table 1: Valence descriptives by condition

Condition	Stories	Turns	Author 1 M	Author 1 SD	Author 2 M	Author 2 SD
Human-AI	87	783	-0.083	0.404	0.004	0.409
Human-Human	38	342	-0.210	0.388	-0.172	0.380
AI-AI	40	360	0.088	0.325	0.073	0.326



5 Alignment Analysis

5.1 Within-Condition Asymmetry Models

For each condition, we fit a single mixed model with **predictor** $\times$ **direction** interaction on standardized valence. This properly tests whether alignment strength differs by direction within each condition.

```
##
## === Human-AI ===
## N = 1566 observations from 87 stories
##
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: response_z ~ predictor_z * alignment_direction + (1 + predictor_z |
## conversation_id)
## Data: df_cond
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 4214.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.68705 -0.71590  0.00875  0.71910  2.74533
##
## Random effects:
## Groups      Name                Variance Std.Dev. Corr
## conversation_id (Intercept) 0.000000 0.00000
## predictor_z      predictor_z 0.007926 0.08903   NaN
## Residual              0.846780 0.92021
## Number of obs: 1566, groups: conversation_id, 87
```

```

##
## Fixed effects:
##
##               Estimate Std. Error      df t value
## (Intercept)    1.351e-17  2.325e-02  1.476e+03  0.000
## predictor_z     1.983e-01  2.645e-02  8.600e+01  7.497
## alignment_direction1 -7.073e-18  2.325e-02  1.476e+03  0.000
## predictor_z:alignment_direction1  5.160e-02  2.466e-02  1.476e+03  2.092
##
##               Pr(>|t|)
## (Intercept)    1.0000
## predictor_z     5.45e-11 ***
## alignment_direction1  1.0000
## predictor_z:alignment_direction1  0.0366 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) prdct_ algn_1
## predictor_z 0.000
## algnmnt_dr1 0.000  0.000
## prdctr_z:_1 0.000  0.000  0.000
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
##
##
## --- Slopes by Direction ---
## alignment_direction predictor_z.trend      SE df lower.CL upper.CL
## A1_to_A2            0.250 0.0362 287    0.1787    0.321
## A2_to_A1            0.147 0.0362 287    0.0755    0.218
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## --- Asymmetry Test (A1→A2 vs A2→A1) ---
## contrast            estimate      SE df t.ratio p.value
## A2_to_A1 - A1_to_A2   -0.103 0.0493 1390  -2.092  0.0366
##
## Degrees-of-freedom method: kenward-roger
##
## === Human-Human ===
## N = 684 observations from 38 stories
##
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: response_z ~ predictor_z * alignment_direction + (1 + predictor_z |
## conversation_id)
## Data: df_cond
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 1849.4
##
## Scaled residuals:
##      Min      1Q    Median      3Q      Max
## -3.01487 -0.72589  0.00344  0.70538  2.54437

```

```

##
## Random effects:
##   Groups      Name      Variance Std.Dev.  Corr
## conversation_id (Intercept) 0.000e+00 0.000e+00
## predictor_z      1.884e-18 1.372e-09  NaN
## Residual          8.554e-01 9.249e-01
## Number of obs: 684, groups: conversation_id, 38
##
## Fixed effects:
##               Estimate Std. Error      df t value
## (Intercept)      9.165e-18  3.536e-02 6.800e+02  0.000
## predictor_z      2.068e-01  3.751e-02 6.800e+02  5.512
## alignment_direction1 2.615e-17  3.536e-02 6.800e+02  0.000
## predictor_z:alignment_direction1 -2.424e-02  3.751e-02 6.800e+02 -0.646
##               Pr(>|t|)
## (Intercept)          1.000
## predictor_z          5.03e-08 ***
## alignment_direction1 1.000
## predictor_z:alignment_direction1 0.518
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) prdct_ algn_1
## predictor_z 0.000
## algnmnt_dr1 0.000 0.000
## prdctr_z:_1 0.000 0.000 0.000
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
##
##
## --- Slopes by Direction ---
## alignment_direction predictor_z.trend      SE df lower.CL upper.CL
## A1_to_A2              0.183 0.053 139   0.0776   0.287
## A2_to_A1              0.231 0.053 139   0.1261   0.336
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## --- Asymmetry Test (A1→A2 vs A2→A1) ---
## contrast      estimate      SE df t.ratio p.value
## A2_to_A1 - A1_to_A2  0.0485 0.075 606   0.646 0.5183
##
## Degrees-of-freedom method: kenward-roger
##
## === AI-AI ===
## N = 720 observations from 40 stories
##
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: response_z ~ predictor_z * alignment_direction + (1 + predictor_z |
## conversation_id)
## Data: df_cond

```

```

## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 1955.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.45939 -0.67072  0.04295  0.68495  2.65544
##
## Random effects:
##   Groups             Name             Variance Std.Dev. Corr
##   conversation_id (Intercept) 0.00000   0.0000
##                   predictor_z 0.04563   0.2136   NaN
##   Residual                0.83793   0.9154
## Number of obs: 720, groups: conversation_id, 40
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)    3.280e-17  3.411e-02 6.770e+02  0.000
## predictor_z     1.306e-01  4.950e-02 3.900e+01  2.639
## alignment_direction1 -2.690e-17  3.411e-02 6.770e+02  0.000
## predictor_z:alignment_direction1 3.189e-02  3.618e-02 6.770e+02  0.881
##
##              Pr(>|t|)
## (Intercept)    1.0000
## predictor_z     0.0119 *
## alignment_direction1 1.0000
## predictor_z:alignment_direction1 0.3784
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) prdct_ algn_1
## predictor_z 0.000
## algnmnt_dr1 0.000  0.000
## prdctr_z:_1 0.000  0.000  0.000
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
##
##
## --- Slopes by Direction ---
## alignment_direction predictor_z.trend      SE    df lower.CL upper.CL
## A1_to_A2              0.1625 0.0613 90.2   0.0407   0.284
## A2_to_A1              0.0987 0.0613 90.2  -0.0231   0.221
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## --- Asymmetry Test (A1→A2 vs A2→A1) ---
## contrast              estimate      SE  df t.ratio p.value
## A2_to_A1 - A1_to_A2 -0.0638 0.0724 638  -0.881  0.3784
##
## Degrees-of-freedom method: kenward-roger

```

5.2 Summary of Within-Condition Asymmetry

Table 2: Within-Condition Asymmetry: Does alignment differ by direction?

Condition	(A1→A2)	(A2→A1)	Diff	SE	t	p	Interpretation
Human-AI	0.250	0.147	-0.103	0.049	-2.092	0.037	A2→A1 stronger *
Human-Human	0.183	0.231	0.048	0.075	0.646	0.518	No asymmetry
AI-AI	0.163	0.099	-0.064	0.072	-0.881	0.378	No asymmetry

6 Cross-Condition Comparison

6.1 Full Model: Condition $\times$ Direction $\times$ Predictor

Test whether asymmetry patterns differ across conditions using a three-way interaction.

```
## === Full Model: predictor  $\times$  direction  $\times$  condition ===

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: response_z ~ predictor_z * alignment_direction * condition +
## (1 + predictor_z | conversation_id)
## Data: df_std
## Control: lmerControl(optimizer = "bobyqa")
##
## REML criterion at convergence: 8033.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.01559 -0.71811  0.00948  0.71683  2.69805
##
## Random effects:
##   Groups             Name             Variance Std.Dev. Corr
##   conversation_id (Intercept) 0.000000  0.0000
##                   predictor_z 0.01329   0.1153   NaN
##   Residual                0.84828   0.9210
## Number of obs: 2970, groups:  conversation_id, 165
##
## Fixed effects:
##
##              Estimate Std. Error      df
## (Intercept)   -1.076e-17  1.814e-02  2.796e+03
## predictor_z     1.785e-01  2.151e-02  1.620e+02
## alignment_direction1 -2.378e-17  1.814e-02  2.796e+03
## condition1      3.100e-18  2.257e-02  2.796e+03
## condition2      2.570e-18  2.724e-02  2.796e+03
## predictor_z:alignment_direction1 1.975e-02  1.924e-02  2.796e+03
## predictor_z:condition1 1.971e-02  2.677e-02  1.620e+02
## predictor_z:condition2 2.821e-02  3.232e-02  1.620e+02
## alignment_direction1:condition1 -7.370e-18  2.257e-02  2.796e+03
## alignment_direction1:condition2 2.336e-18  2.724e-02  2.796e+03
## predictor_z:alignment_direction1:condition1 3.185e-02  2.394e-02  2.796e+03
## predictor_z:alignment_direction1:condition2 -4.399e-02  2.890e-02  2.796e+03
##
##              t value Pr(>|t|)
## (Intercept)      0.000    1.000
## predictor_z      8.300 3.88e-14 ***
```

```

## alignment_direction1          0.000    1.000
## condition1                    0.000    1.000
## condition2                    0.000    1.000
## predictor_z:alignment_direction1 1.027    0.305
## predictor_z:condition1          0.736    0.463
## predictor_z:condition2          0.873    0.384
## alignment_direction1:condition1 0.000    1.000
## alignment_direction1:condition2 0.000    1.000
## predictor_z:alignment_direction1:condition1 1.331    0.183
## predictor_z:alignment_direction1:condition2 -1.522    0.128
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) prdct_ algn_1 cndtn1 cndtn2 pr_:_1 prd_:1 prd_:2 al_1:1
## predictor_z  0.000
## algnmnt_dr1  0.000  0.000
## condition1  -0.362  0.000  0.000
## condition2   0.171  0.000  0.000 -0.431
## prdctr_z:_1  0.000  0.000  0.000  0.000  0.000
## prdctr_z:c1  0.000 -0.362  0.000  0.000  0.000  0.000
## prdctr_z:c2  0.000  0.171  0.000  0.000  0.000  0.000 -0.431
## algnmnt_1:1  0.000  0.000 -0.362  0.000  0.000  0.000  0.000  0.000
## algnmnt_1:2  0.000  0.000  0.171  0.000  0.000  0.000  0.000  0.000 -0.431
## prdct_:_1:1  0.000  0.000  0.000  0.000  0.000 -0.362  0.000  0.000  0.000
## prdct_:_1:2  0.000  0.000  0.000  0.000  0.000  0.171  0.000  0.000  0.000
##      al_1:2 p_:_1:1
## predictor_z
## algnmnt_dr1
## condition1
## condition2
## prdctr_z:_1
## prdctr_z:c1
## prdctr_z:c2
## algnmnt_1:1
## algnmnt_1:2
## prdct_:_1:1  0.000
## prdct_:_1:2  0.000 -0.431
## optimizer (bobyqa) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')

```

6.2 Condition Comparison of Asymmetry

```

## === Alignment Slopes by Condition × Direction ===
##      condition  alignment_direction predictor_z.trend      SE  df lower.CL upper.CL
## Human-AI      A1_to_A2              0.2499 0.0370 505  0.17710  0.323
## Human-Human  A1_to_A2              0.1825 0.0560 505  0.07242  0.293
## AI-AI        A1_to_A2              0.1625 0.0546 505  0.05521  0.270
## Human-AI      A2_to_A1              0.1467 0.0370 505  0.07390  0.219
## Human-Human  A2_to_A1              0.2310 0.0560 505  0.12091  0.341
## AI-AI        A2_to_A1              0.0987 0.0546 505 -0.00857  0.206
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

```



```

##
## === Asymmetry Contrasts Within Each Condition ===
## condition = Human-AI:
## contrast          estimate      SE   df t.ratio p.value
## A2_to_A1 - A1_to_A2 -0.1032 0.0494 2634 -2.090 0.0367
##
## condition = Human-Human:
## contrast          estimate      SE   df t.ratio p.value
## A2_to_A1 - A1_to_A2  0.0485 0.0747 2634  0.649 0.5164
##
## condition = AI-AI:
## contrast          estimate      SE   df t.ratio p.value
## A2_to_A1 - A1_to_A2 -0.0638 0.0728 2634 -0.876 0.3811
##
## Degrees-of-freedom method: kenward-roger
##
## === Does Asymmetry Differ Across Conditions? ===
## condition_revpairwise alignment_direction_revpairwise estimate      SE
## (Human-Human) - (Human-AI) A2_to_A1 - A1_to_A2          0.1517 0.0895
## (AI-AI) - (Human-AI)      A2_to_A1 - A1_to_A2          0.0394 0.0880
## (AI-AI) - (Human-Human)   A2_to_A1 - A1_to_A2         -0.1123 0.1040
## df t.ratio p.value
## 2634  1.694 0.0904
## 2634  0.448 0.6541
## 2634 -1.076 0.2819
##
## Degrees-of-freedom method: kenward-roger

```

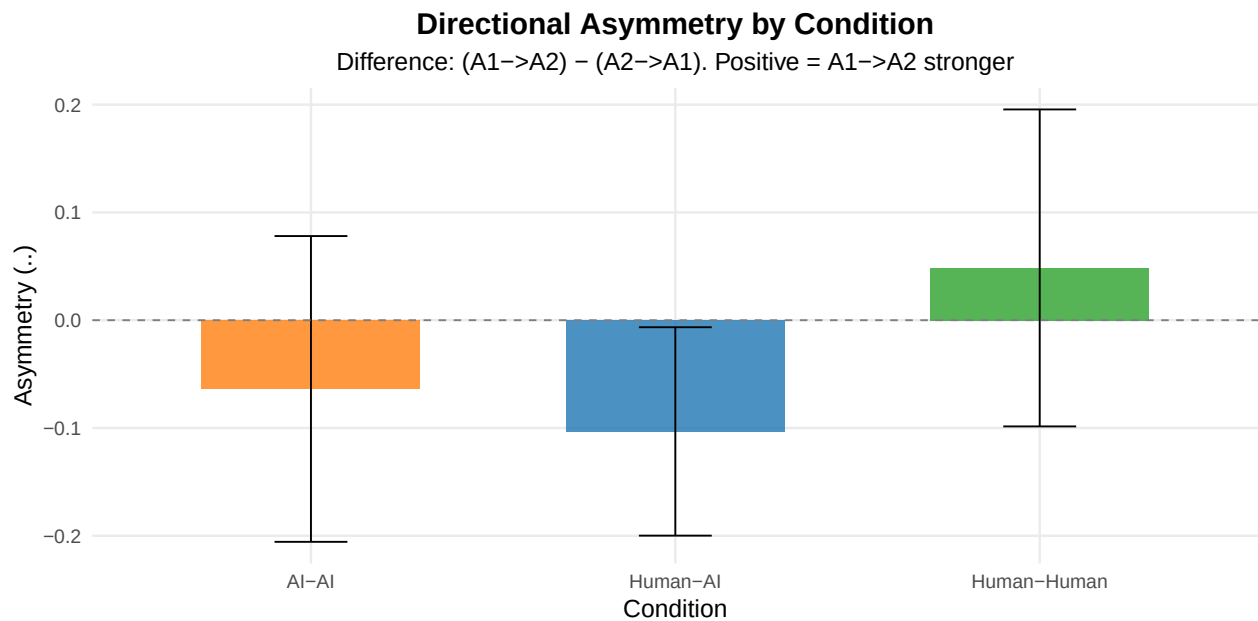
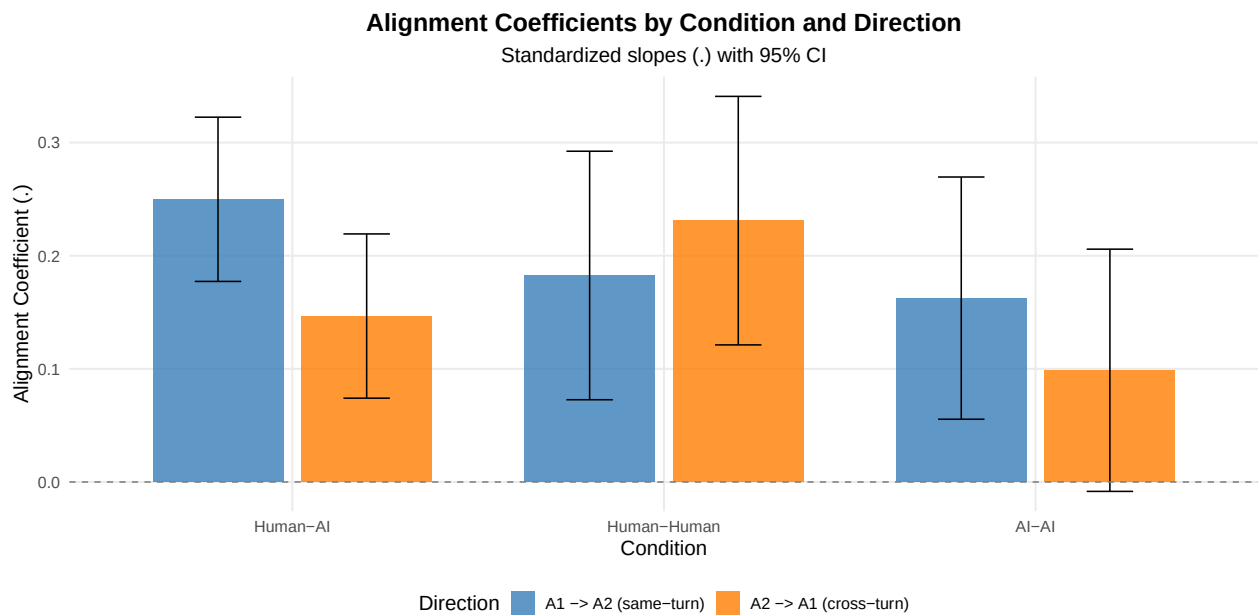
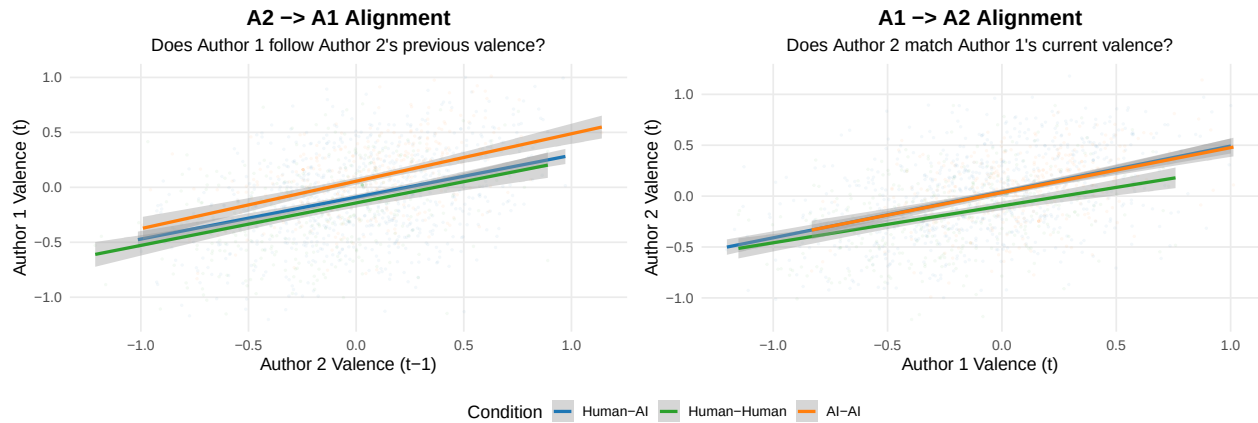
6.3 Pairwise Condition Contrasts for Each Direction

```

## === Condition Contrasts for A1→A2 Direction ===
## alignment_direction = A1_to_A2:
## contrast          estimate      SE   df t.ratio p.value
## (Human-AI) - (Human-Human) 0.0674 0.0672 505  1.003 0.9494
## (Human-AI) - (AI-AI)       0.0873 0.0660 505  1.324 0.5586
## (Human-Human) - (AI-AI)    0.0200 0.0783 505  0.256 1.0000
##
## alignment_direction = A2_to_A1:
## contrast          estimate      SE   df t.ratio p.value
## (Human-AI) - (Human-Human) -0.0843 0.0672 505 -1.256 0.6294
## (Human-AI) - (AI-AI)       0.0479 0.0660 505  0.726 1.0000
## (Human-Human) - (AI-AI)    0.1323 0.0783 505  1.690 0.2748
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests

```

7 Visualization



8 Summary

Table 3: Summary: Alignment slopes and asymmetry by condition

Condition	A1→A2	A2→A1	Asymmetry (p)	Interpretation
Human-AI	0.250	0.147	-0.103 (0.037) *	A2→A1 stronger
Human-Human	0.183	0.231	0.048 (0.518) ns	No asymmetry
AI-AI	0.163	0.099	-0.064 (0.378) ns	No asymmetry

9 Key Findings

Based on the analysis above:

1. **Within-condition asymmetry:** Each condition is tested for directional asymmetry using a predictor $\times$ direction interaction within a single model. A significant interaction indicates that alignment strength differs between the A1→A2 and A2→A1 directions.
2. **Cross-condition comparison:** The three-way interaction (predictor $\times$ direction $\times$ condition) tests whether the *pattern* of asymmetry differs across conditions. A significant three-way interaction means some conditions show asymmetry while others do not, or asymmetry is in opposite directions.
3. **Interpretation for Human-AI:** In the human-ai condition, if A1→A2 (User→AI) is significantly stronger than A2→A1 (AI→User), this indicates the AI adapts to human valence but humans do not follow AI valence.
4. **Comparison insights:**
 - If Human-Human shows no asymmetry but Human-AI does, this suggests asymmetric accommodation is specific to human-AI interaction.
 - If AI-AI shows strong symmetric alignment, it serves as a baseline for what “full mutual alignment” looks like.

Analysis completed: 2026-04-07 18:49:09.198643